

CSC-355 Homework #6

unification

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This homework is an opportunity for you to practice the steps in the unification algorithm.

Problem 1: Unification Basics

Do the correct transformation on each of the following equation sets or say why the unification algorithm would fail.

- (a) $\{x \stackrel{?}{=} x\}$
- (b) $\{f(X, Y, Z) \stackrel{?}{=} g(X, Y, Z)\}$
- (c) $\{f(X, Y) \stackrel{?}{=} Z\}$
- (d) $\{X \stackrel{?}{=} g(f(Y, X), Z)\}$
- (e) $\{f(X, Y) \stackrel{?}{=} f(X, Y, Z)\}$
- (f) $\{Y \stackrel{?}{=} f(Z), g(Y, Y) \stackrel{?}{=} X\}$

Problem 2: Unification

Show the steps of the unification algorithm on and give the mgu (if it exists) for the following equation sets:

- (a) $\{f(X, h(Z)) \stackrel{?}{=} f(h(Z), h(h(Y)))\}$
- (b) $\{f(X, g(Y)) \stackrel{?}{=} f(h(Z, Z), X)\}$
- (c) $\{f(X, h(Y)) \stackrel{?}{=} f(Y, X)\}$